

BEYOND OUR EMOTIONS and lack of rationality, we have an entirely different set of issues that are hard-wired into our brains: *Cognition-based errors*.

The way our brains have evolved over the past two million years or so has led to issues inherent in our wetware. Decisions about money were not a big issue on the Serengeti plains, but *avoiding being eaten by lions* was.

In medicine, drugs prescribed for purposes other than those they were created for are called “off-label” uses. Similarly, our brains evolved for use in a very different environment than the modern world. As a result, a predictable set of errors affects how we think and act.

The technical term for this is the “evolutionary mismatch hypothesis.” This theory states that traits that were once advantageous in the wild become maladaptive in rapidly changing environments. It’s what happens when attributes created over millions of years are suddenly thrust into a very different ecosystem. It also perfectly describes today’s investors, saddled with biological traits that cannot adapt fast enough to modern capital markets.

This leads to a few unfortunate tendencies that get in our way:

- We see patterns where none exist.
- We have difficulty conceptualizing long arcs of time.
- We do not experience exponential numbers in nature, so compounding presents an instinctive challenge to us intellectually.
- We selectively perceive and recall that which agrees with our preexisting expectations while ignoring things that contradict our beliefs.
- Our ego makes us believe we are much more capable than we really are; we forget our failures while overemphasizing our successes.
- We are also threat-biased—our brains are better at processing bad